

Time Series Analysis of Daily Bitcoin Closing Prices

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Abstract

With the increasing adoption of cryptocurrencies, high-frequency trading (HFT) and algorithmic strategies have become key components of financial markets. In this project, we aim to perform a time series analysis of Bitcoin (BTC) pricing over a period of 11 years from 2014 till present at a high-frequency resolution, leveraging Open-High-Low-Close (OHLC) data collected from Yahoo finance. The objective is to explore market trends, identify patterns, and use time series modelling techniques to predict price movements in Bitcoin. By analyzing these aspects, we aim to provide valuable insights into market efficiency, and algorithmic trading strategies.

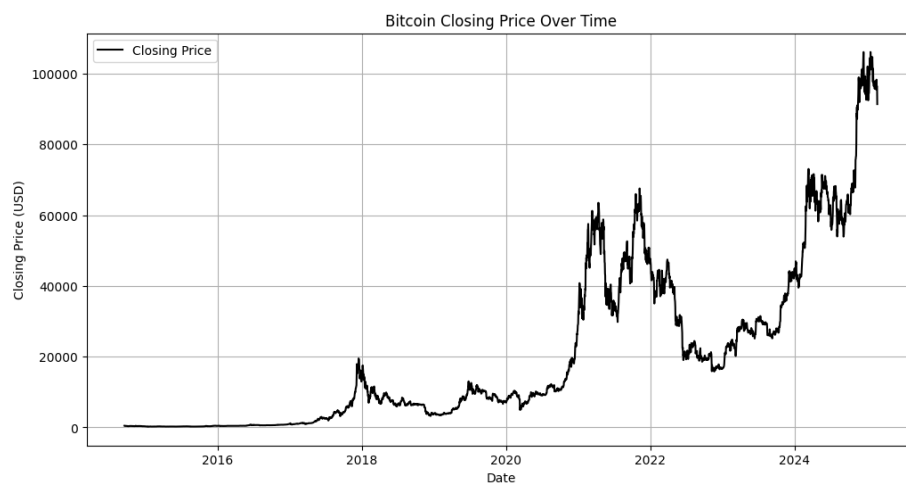


Figure 1: Bitcoin Closing Price (in \$) Over Time

Link to Dataset Source: [Yahoo Finance](#)